Supplier Selection for Vietnamese Arabica Green Coffee

1. Introduction

Supplier selection is a key strategic decision that affects overall supply chain performance. Businesses face growing supply chain risks, such as extreme weather events, stricter environmental and food safety regulations, and shifts in trade policies. Relying on a single traditional criterion, such as purchasing price, is therefore unlikely to provide a sustainable competitive advantage for exporting firms.

This case study examines eight hypothetical suppliers (S01-S08) of Vietnamese fully washed Arabica green coffee beans, with Screen 16 assumed as a common specification. An exporting company purchases coffee from domestic suppliers for export and compares suppliers on a common 20-metric-tonne (MT) reference quantity. Supplier prices are set on an EXW basis, while estimated buyer-paid transport and storage costs are added separately.

The objective is to identify the preferred eligible supplier under the Base weighting scenario and assess whether the recommendation changes when the company places greater emphasis on cost or traceability. The analysis uses mandatory screening, the Weighted Sum Model (WSM), the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS), and a three-scenario sensitivity analysis. All supplier data are simulated, so this is an analytical demonstration rather than an assessment of real businesses.

2. Methodology

2.1. Proposed Supplier Selection Criteria

The proposed supplier selection framework draws on criteria discussed in previous studies and adapts them to the procurement of Vietnamese Arabica green coffee. Six factors were considered, including Purchasing Price, Logistics Cost, Product Quality, On-time Delivery, Traceability, and Certificate (Table 1).

Table 1. Supplier Selection Criteria

Criterion	Measurement and direction	Source
Purchasing Price	EXW price (USD/MT); lower is better	Nguyen et al. (2021); Wang et al. (2022)
Logistics Cost	Estimated transportation cost from the supplier's warehouse to the consolidation point, plus storage fees (USD/MT); lower is better.	Nguyen et al. (2021); Wang et al. (2022)
Product Quality	Full defects per 300g; lower is better	Dickson (1966); Wang et al. (2022); Rahmawati & Salimi (2022); Dania et al. (2023)
On-time Delivery	On-time deliveries/total deliveries × 100 (%); higher is better	Dickson (1966); Rahmawati & Salimi (2022)
Traceability	Traceability evidence score 0-100; higher is better	Ali et al. (2017); Islam et al. (2021); Rahmawati & Salimi (2022)

Criterion	Measurement and direction	Source
Certificate	Binary indicator (1/0) of compliance with the buyer-defined documentation requirement	Banaeian et al. (2015); Wang et al. (2022)

Source: Author's synthesis

The evaluation follows a two-stage approach. First, suppliers must satisfy the buyer-defined minimum requirements before being considered for further evaluation. These requirements are presented in Table 2. The processing method, screen size, and packaging are assumed to meet the same product specifications across all eight hypothetical suppliers.

Table 2. Mandatory buyer-defined screening conditions

Requirement	Description
Moisture content	9%-13%
Full defects	≤23 full defects per 300g
Certificate	1 = Meets the buyer-defined documentation requirement; 0 = Does not meet the requirement

Source: Author's assumptions

Second, for the quantitative evaluation, Purchasing Price and Logistics Cost are combined into Delivered Cost, providing a common cost basis for comparing suppliers. In this study, product quality is measured by the number of full defects per 300g of green coffee, but in practice, defect counts alone are insufficient to assess coffee quality. The certificate is treated solely as a mandatory requirement rather than a compensatory ranking criterion. Traceability is measured using five project-defined binary indicators: production origin documentation (T1), lot identification (T2), lot-origin linkage (T3), lot-level movement records (T4), and documentation verification (T5). These indicators were developed with reference to selected GS1 traceability principles (GS1, 2017). Each indicator contributes 20 points, producing a total score ranging from 0 to 100.

Following screening, eligible suppliers were ranked using four criteria: Delivered Cost, Product Quality, On-time Delivery, and Traceability. Two multi-criteria decision-making methods were applied. The WSM aggregates normalized criterion values using assigned weights, while the TOPSIS evaluates suppliers according to their distances from positive and negative ideal solutions (Abhilasha et al., 2024). For WSM, Delivered Cost was normalized using min(x)/x, defect counts using reversed min-max normalization, and On-time Delivery and Traceability using x/max(x). TOPSIS applied vector normalization, treating Delivered Cost and defects as cost criteria. Both methods used the same eligible suppliers and criterion weights.

The Base scenario assigned 40% weight to Delivered Cost and 20% to each remaining criterion. Two alternative scenarios, Cost-focused and Traceability-focused, were introduced to examine the sensitivity of supplier rankings to changes in buyer priorities. All weights were defined as case-study assumptions rather than derived from an expert survey or Analytic Hierarchy Process (AHP).

The analysis was implemented in Google Colab using Python, primarily pandas and NumPy. The accompanying notebook and Excel workbooks provide the detailed calculations and results.

3. Results and Discussion

3.1. Supplier Screening Results

Six of eight suppliers meet all three buyer-defined requirements. S02 records 25 full defects per 300g, exceeding the maximum of 23; S08 has Certificate = 0. Both are excluded before scoring. S01, S03, S04, S05, S06 and S07 proceed to WSM and TOPSIS.

3.2. WSM and TOPSIS Results

Figure 1 shows the Base-scenario rankings between the WSM and TOPSIS methods. Both methods place S01 first and S05 second, while S07 ranks last among eligible suppliers. The principal disagreement concerns S03, S04, and S06.

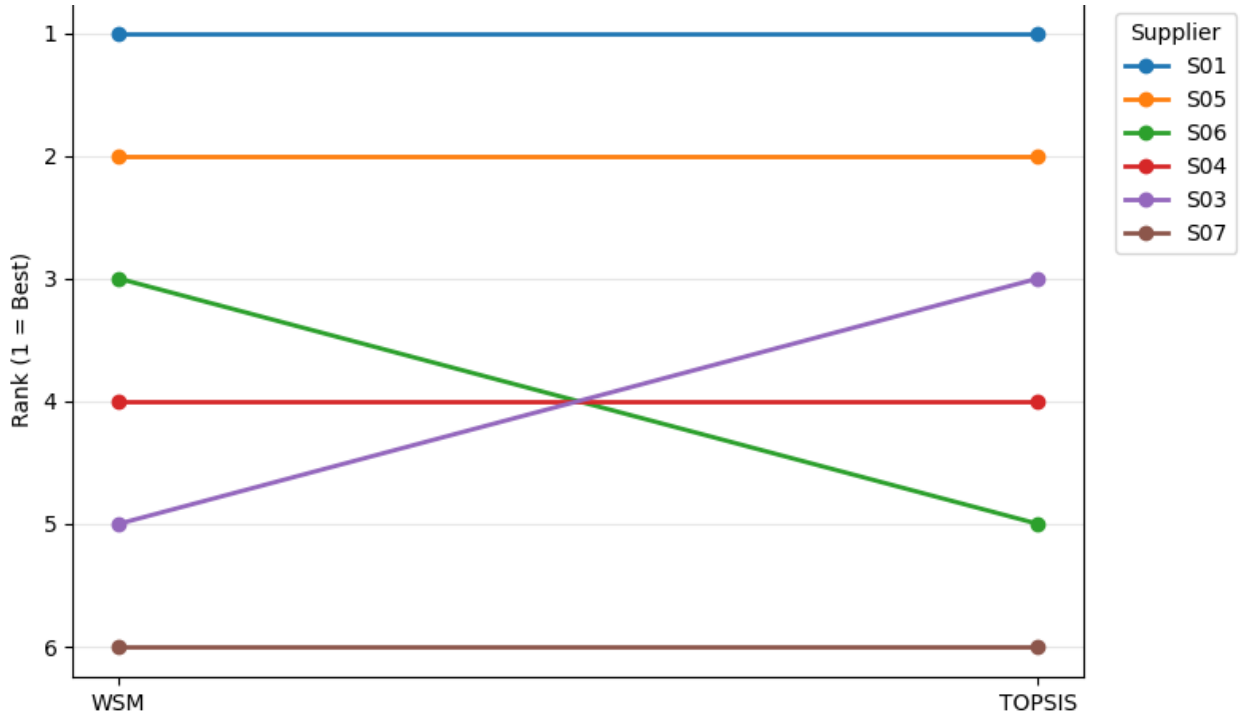


Figure 1. Supplier Ranking Comparison: WSM vs TOPSIS (rank 1 = best)

Source: Author's calculations

S01 leads because its favorable physical defect result (3 full defects/300g) and assumed on-time delivery performance (90%) offset its higher estimated cost relative to S05. S05 has the lowest modeled Delivered Cost among eligible suppliers at USD 6,087.10/MT, compared with USD 6,220.50/MT for S01. The S01 premium is USD 133.40/MT (approximately 2.2%), or USD 2,668 for the 20-MT reference quantity. Both suppliers have a simulated Traceability Score of 100/100.

Table 3. Comparison of the two leading suppliers

Measure	**S01**	**S05**
Delivered Cost (USD/MT)	6,220.50	6,087.10
Full defects / 300g	3	5
On-time Delivery	90%	86%
Traceability Score	100/100	100/100
Base WSM / TOPSIS rank	1 / 1	2 / 2

Source: Author's calculations

The middle rankings depend on the aggregation method. S06 is third under WSM but fifth under TOPSIS; S03 moves from fifth to third. S06 has a substantially lower modeled Delivered Cost (USD 6,122.40/MT versus USD 7,006.20/MT for S03) and higher OTD and traceability scores, whereas S03 has fewer full defects (8 versus 12). WSM and TOPSIS combine these trade-offs differently, therefore, the ranking discrepancy should not be interpreted as evidence that one method is universally superior.

3.3. Sensitivity Analysis Results

Table 4 compares the six eligible suppliers under the three weighting scenarios. The two leading suppliers are unchanged in every method-scenario combination.

Table 4. Supplier ranks across weighting scenarios

Supplier	Base WSM	Base TOPSIS	Cost WSM	Cost TOPSIS	Trace WSM	Trace TOPSIS
S01	1	1	1	1	1	1
S05	2	2	2	2	2	2
S03	5	3	5	4	5	3
S04	4	4	4	5	4	5
S06	3	5	3	3	3	4
S07	6	6	6	6	6	6

Note: Cost = Cost-focused; Trace = Traceability-focused. Rank 1 is best. S02 and S08 remain excluded in all scenarios.

Source: Author's calculations

S01 consistently ranks first and S05 second under both WSM and TOPSIS across all three weighting scenarios. WSM rankings remain unchanged, while TOPSIS produces some changes among the middle-ranked suppliers. S06 moves from fifth in the Base scenario to third in the Cost-focused scenario and fourth in the Traceability-focused scenario. S03 ranks third, fourth, and third, while S04 ranks fourth, fifth, and fifth, respectively. These results suggest that S01 and S05 are relatively stable choices under the tested scenarios, although different weights, quality measures, or input data could change the rankings.

4. Procurement Recommendation

Based on the analysis, S01 is recommended as the preferred supplier for the hypothetical purchase, with S05 as the backup option. S01 ranks first under both WSM and TOPSIS and maintains this position across all three weighting scenarios. As the analysis uses simulated data, an actual purchasing decision would require verification of supplier quotations, product samples, supporting documentation, and logistics arrangements.

5. Limitations

This study has several limitations. The data describe eight hypothetical suppliers, so a company applying the framework would need to collect real supplier data. The screening thresholds and criterion weights were chosen for this case rather than elicited from stakeholders, and the criteria

themselves would differ from one company to another. Finally, the sensitivity analysis varies the weights across only three scenarios. It does not test uncertainty in the supplier inputs, other normalization methods, or other buyer preferences. The results therefore show how the framework works, not how real suppliers perform.

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